Payload-Aware Two-Tier UAV Route Planning for Last-Mile Delivery

Review Report – I

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Presentation Outline

1. Introduction
2. Literature Review
3. System Design
4. Implementation and Results
5. Project Management
6. Conclusion

Background and Motivation

- UAVs are an effective platform for last-mile delivery where ground transport is impractical: remote clinics, emergency drug drops, and fragile diagnostic sample transport.
- Every flight must balance three **conflicting objectives**: total distance, battery energy use, and exposure to high-risk flight segments.
- Existing routing systems treat all deliveries **identically** — a fragile medical device and a robust parcel travel the *same* route.
- Payload metadata (fragility, urgency, payload weight) encodes routing priorities, but **no system exploits this at mission time**.

Core Motivation

Payload-blind UAV routing pipelines cannot adapt to cargo-specific constraints. A two-tier architecture that bridges payload semantics directly to Pareto route selection addresses this structural gap.

Real-World Cost of Payload-Blind Routing

Swiss Post – Matternet M2 Crashes (Zürich, Jan & May 2019) [8]

Drones carrying clinical laboratory specimens flew **fixed, payload-invariant routes**. Post-incident analysis confirmed identical corridors regardless of cargo type — no routing adjustment for payload fragility.

Vibration & Turbulence Degradation Studies [9, 10, 11]

Controlled lab work shows drone-transported medicines degrade measurably: insulin fibrillation, biologic protein denaturation, elevated haemolysis in blood samples on high-vibration segments. Altitude variation and turbulence raise lab rejection rates.

Zipline Rwanda Blood-Delivery Network (up to 80 km/charge) [12]

Heavier blood-product payloads draw more hover/lift power and shorten range, yet the system flies the same fixed route regardless of payload weight — no energy-efficient detour for heavy cargo, no direct short route for urgent light payload.

Root cause in every case: the planner does not know what is carried, so it cannot prefer smoother corridors for fragile payloads or shortest paths for time-critical medicine.

Problem Statement

Problem

Existing UAV routing systems apply **static, cargo-agnostic objective weights**, resulting in **suboptimal or unsafe routes** for payloads with differing fragility, urgency, and payload weight.

- Given payload metadata $(F, W, U, P_{\text{lvl}})$ with $F \in \{1 \dots 5\}$, $W \in \{1 \dots 3\}$, $U \in \{1 \dots 3\}$, $P_{\text{lvl}} \in \{1 \dots 5\}$, and airspace graph $G = (V, E)$, select route p^* from the Pareto front that best satisfies payload requirements — no expert re-tuning, no re-running the search.
- Single-objective routing collapses all objectives into one scalar and cannot model their three-way conflict.
- No existing system maps payload description to Pareto point selection via an explicit algebraic formula.
- Constraints: selection must be deterministic, reproducible, and finish within 1 s of payload tag entry.

Review 0: Feedback & Actions

Panel Comment	Resolution in Current Work
"No correlation between Tier 1 and Tier 2 factors"	Explicit algebraic mapping introduced: Fragility F scales w_{risk}; Urgency U scales w_{dist}; Payload weight W scales w_{energy}. Every Tier 2 tag maps to a Tier 1 objective via a closed formula.
"Criteria are not considered to map together"	The formula $w_{\text{risk}} = (F{\times}2) \,/\, (F{\times}2 + U{\times}3 + W{\times}2)$ (and equivalents) is a continuous function — all four tags simultaneously influence all three weights; changing any single tag shifts all three weights proportionally.
"Application of Pareto needs rethinking"	Extreme-point selection **replaced** by utility scoring over the *full* Pareto front: $\text{score}(p_i) = w_d(1-\tilde{d}_i) + w_\varepsilon(1-\tilde{\varepsilon}_i) + w_\rho(1-\tilde{\rho}_i)$. Mixed-priority payloads select *non-extreme interior points* that no single-objective baseline can produce.
"Complexity only in Tier 2; more than 2 objectives"	Tier 1 manages three-objective NAMOA* graph-search complexity; Tier 2 manages metadata-to-weight mapping and full-front utility scoring. Both tiers carry distinct computational roles — neither reduces to a single-objective problem.

Project Objectives

1. To build a **multi-objective A* engine (Tier 1)** that constructs a Pareto-optimal route set over an airspace graph, balancing distance, energy consumption, and flight risk simultaneously.
2. To design an **algebraic bridge** that translates payload metadata tags (fragility F, payload weight W, urgency U, priority P_{lvl}) into normalised routing objective weights with no expert tuning.
3. To implement a **Tier 2 payload-driven selector** that scores every Pareto route using the derived weights and returns the utility-maximising route p^*.
4. To evaluate the pipeline using **simulation benchmarks** on 50, 200, and 500-node graphs against single-objective A* and MOPSO baselines, using hypervolume, Pareto spread, IGD, and Tier 2 selection accuracy.
5. To validate the system on **real drone hardware** (Raspberry Pi 5 + Pixhawk) through ArduPilot SITL simulation and outdoor flight tests.

Scope and Expected Outcomes

Included

- Payload-aware UAV route selection for single-drone delivery
- Two-tier pipeline: Pareto front construction + utility-based Pareto point selection
- Software simulation on OSM graphs
- ArduPilot SITL + RPi 5 + Pixhawk hardware integration and outdoor flight tests
- Benchmark against single-objective A* and MOPSO

Excluded / Constraints

- Multi-UAV fleet management and coordination
- Real-time dynamic obstacle avoidance during flight
- Commercial deployment and regulatory compliance
- Weather-based in-flight re-routing
- Deep-learning-based risk estimation

Literature Review

Author / Year	Method	Key Contribution	Limitation / Research Gap
Chokkalingam & Ganesan (2026) [1]	Pareto-entropy PSO (POPSO) with dynamic parameter adaptation	Entropy-guided archiving improves front diversity; adaptable to real-time environmental changes	Final route still a manually re-tuned weighted sum per mission; payload metadata never incorporated — fragile samples and emergency medicine get the same hand-tuned selection
Tsoutsouras et al. (2020) [7]	Mass-aware flight-speed and energy-model adaptation for multi-drop trajectories	Reduces energy consumption vs. mass-agnostic planning across multi-stop routes	Continuous mass used only inside the energy/speed model itself; builds no multi-objective Pareto front; discrete payload metadata (fragility, urgency) plays no role in route selection
Ren et al. (2019) [2]	Crowding-distance NSGA-II (CDNSGA-II) over octree urban environment	Well-spread, collision-free Pareto-optimal paths in urban airspace	Crowding distance encodes only geometric front spread; urgency, fragility, and payload weight play no role in selection

Research Gap and Proposed Contribution

Three Identified Gaps

- **Gap 1:** No payload-aware route *selection* — all planners apply a fixed geometric criterion independent of payload type, or no selection mechanism at all
- **Gap 2:** No front reuse — full multi-objective search re-run whenever mission context changes
- **Gap 3:** No algebraic bridge — no formalised mapping from payload description to the routing trade-off it implies

Proposed Contribution (closes all three)

- Tier 2 maps four payload tags directly to Pareto point selection via formula **(Gap 1)**
- Front refreshed periodically; Tier 2 selects per mission against current front — no re-run **(Gap 2)**
- Closed-form bridge: $F, W, U \rightarrow w_{\text{risk}}, w_{\text{dist}}, w_{\text{energy}}$ **(Gap 3)**
- $p^* = \arg\max_{p_i \in \mathcal{P}} \text{score}(p_i)$ — selects non-extreme interior points for mixed payloads

Existing System vs. Proposed System

Criterion	Existing Routing Systems	Proposed Two-Tier System
Routing approach	Single-objective (distance or energy only)	Multi-objective NAMOA* — distance, energy, and risk simultaneously
Payload handling	Cargo-agnostic; all parcels treated identically	Payload tags (F, W, U, P_{lvl}) derive objective weights algebraically
Route selection	Single greedy path or extreme Pareto point; expert re-tunes weights per mission	Utility scoring over full Pareto front; selects non-extreme interior points for mixed payloads
Front reuse	Full search re-run on every mission context change	Front built once per region; Tier 2 selects in <1 s against current front
Hardware validation	Simulation only	ArduPilot SITL + RPi 5 + Pixhawk outdoor flight tests

Proposed Methodology

Payload-Aware Two-Tier UAV Route Planning

System Architecture — Deterministic Pareto Front Construction with Payload-Driven Route Selection

① INPUTS & DATA

Airspace Graph

OpenStreetMap (osmnx) — road/path network converted into the delivery-region airspace graph



Payload Metadata Tags

Fragility 1–5 · Weight 1–3
Urgency 1–3 · Priority 1–5



② TIER 1 — PARETO FRONT CONSTRUCTION

Refreshed periodically per region — not rebuilt per mission

2.1 Edge Cost Assignment

Each edge gets a 3-objective cost vector: distance (km) · energy (battery %) · flight risk (%)

2.2 Multi-Objective A* / Dijkstra

Vector-cost graph search over the airspace graph — deterministic, reproducible front



2.3 Pareto Dominance Pruning

Discards any route dominated on all three objectives simultaneously

2.4 Pareto Front Storage

Persisted set of non-dominated routes — reused across missions within the refresh window



③ TIER 2 — PAYLOAD-DRIVEN SELECTION

Runs per mission — near-instantaneous rule lookup

3.1 Payload Metadata Interpreter

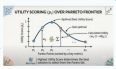
Reads Fragility (F), Weight (W), Urgency (U), Priority (P) tags from operator input

3.2 Weight Derivation Formula

$w_risk = F \cdot 2/\Sigma \quad w_dist = U \cdot 3/\Sigma \quad w_energy = W \cdot 2/\Sigma$ — algebraic bridge to Tier 1 objectives

3.3 Normalisation + Utility Scoring

$score(p_i) = \Sigma\, w_k \cdot (1 - \hat{o}b_{ik})$ computed for EVERY stored Pareto point



3.4 argmax Route Selection

Selects non-extreme points for mixed payloads; priority ≥ 4 amplifies the dominant weight

④ MISSION OUTPUT

Mission File Generation

ArduPilot-compatible .waypoints file from the selected route



Ground-Station Dashboard

Operator enters payload tag once — no manual re-tuning

⑤ UAV CONTROL ↻

MAVLink Upload

Mission file transmitted to the onboard Pixhawk flight controller

SITL → Real Flight

ArduPilot SITL validates first; outdoor flight confirms. Route adherence & latency feedback ↻ informs the next refresh

⑥ OUTPUTS

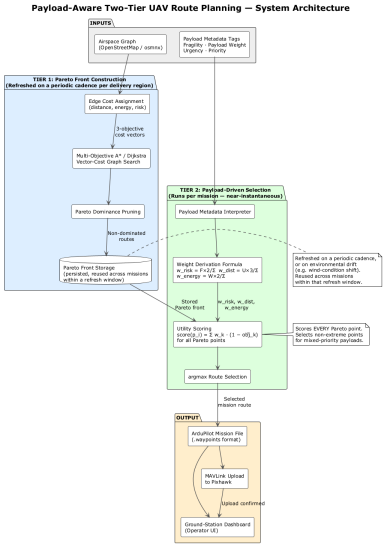
- Payload-Matched Optimal Route
- Ready-to-Fly Mission File
- Reusable Pareto Front — no per-mission re-planning
- Non-Extreme Selection for Mixed Payloads
- SITL + Flight-Validated Pipeline

payload metadata tags — direct to Tier 2 (bypasses Tier 1)

⑦ PERFORMANCE EVALUATION

Benchmark Scenarios	Pareto Front Quality	Tier 2 Selection Accuracy	Computation Time	Baseline Comparison
S1 50n (5%) · S2 200n (20%) · S3 500n (40%)	Hypervolume (HV) · Pareto Spread / IGD	Alignment of selected route vs. payload tag across mission types	Tier 1 pre-computation vs. MOPSO runtime	vs. Single-Objective A* · MOPSO (100 particles, 200 iter) — mean ± std over 30 runs

Proposed System Architecture



System Modules

1. **M1 – Airspace Graph Builder:** Extracts OSM road/path network via `osmnx`; assigns three objective costs to each edge: geodesic distance, energy (distance $\times$ altitude-change factor), and flight risk (simulated threat zones and obstacle density, scored 0–1).
2. **M2 – Multi-Objective A* Engine (Tier 1):** NAMOA* search maintaining a cost vector $\mathbf{g} = [d, \varepsilon, \rho]$ per partial path; node-level label-dominance pruning; stores deterministic Pareto front $\mathcal{P}$ on disk for Tier 2 reuse.
3. **M3 – Payload Metadata Interpreter (Tier 2):** Reads tags $(F, W, U, P_{\text{lvl}})$; derives $w_{\text{risk}}, w_{\text{dist}}, w_{\text{energy}}$ via closed formula; min-max normalises objectives; returns arg max route and generates ArduPilot `.waypoints` mission file.
4. **M4 – Testing and Evaluation:** Benchmarks on S1/S2/S3 graphs vs. single-objective A* and MOPSO (30 runs); metrics: hypervolume, Pareto spread (Δ), IGD, Tier 2 selection accuracy, computation time. Extends to ArduPilot SITL and outdoor flight tests.

Methodology / Workflow

1. Literature review and problem formulation
2. Airspace graph construction (OpenStreetMap + OSMnx)
3. Multi-objective A* (NAMOA*) engine implementation (Tier 1)
4. Algebraic bridge and payload-driven Pareto selector (Tier 2)
5. Software simulation and benchmarking vs. A* and MOPSO baselines
6. Hardware integration (RPi 5 + Pixhawk), ArduPilot SITL simulation, and outdoor flight validation

Evaluation Metrics

- Hypervolume (HV)
- Pareto spread (Δ metric)
- IGD vs. reference front
- Tier 2 selection accuracy
- Tier 1 computation time
- Route adherence (GPS error)

Core Algorithm: Payload-Aware Two-Tier Route Planning

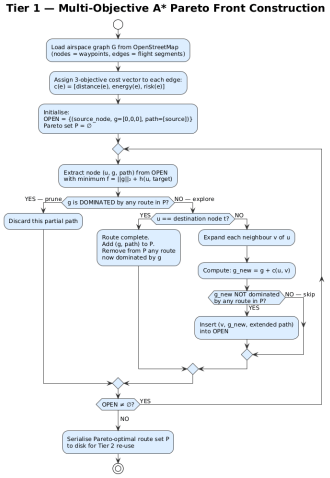
Algorithm Summary

1. **Input:** Graph $G = (V, E)$, payload tags $(F, W, U, P_{\text{lvl}})$, source s, destination d.
2. **Weight derivation (Algebraic Bridge):**
   $r_{\text{raw}}{=}F{\times}2, \quad d_{\text{raw}}{=}U{\times}3, \quad e_{\text{raw}}{=}W{\times}2; \quad$ normalise $\Rightarrow w_{\text{risk}}, w_{\text{dist}}, w_{\text{energy}}$.
   If $P_{\text{lvl}} \geq 4$: amplify dominant weight by $0.1{\times}w_{\max}$, then renormalise to sum to 1.
3. **Tier 1 (NAMOA*):** Run multi-objective A*; maintain cost vectors; apply node-level label-dominance pruning; store Pareto front $\mathcal{P}$ on disk for reuse.
4. **Tier 2 — Utility Scoring:** For each $p_i \in \mathcal{P}$,

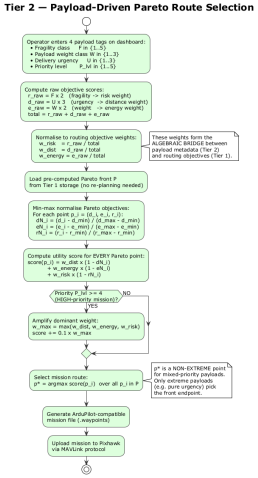
$$\text{score}(p_i) = w_{\text{dist}}(1 - \tilde{d}_i) + w_{\text{energy}}(1 - \tilde{\varepsilon}_i) + w_{\text{risk}}(1 - \tilde{\rho}_i)$$

5. **Output:** $p^* = \arg\max_{p_i \in \mathcal{P}} \text{score}(p_i)$; generate ArduPilot `.waypoints` file; upload via MAVLink.

Activity Flow Diagrams for Two-Tier Architecture



Tier 1: Multi-Objective A* (NAMOA*) Pareto Front Construction



Tier 2: Payload-Driven Pareto Route Selection

Technology Stack and Resources

Category	Selected Technology / Resource
Programming language	Python 3.11
Core libraries	NetworkX, OSMnx, NumPy, matplotlib, SciPy
Communication protocol	MAVLink / ArduPilot `.waypoints` mission format
Simulation environment	ArduPilot SITL (primary validation before real flights)
Map / dataset	OpenStreetMap airspace graph
Development tools	Git, VS Code, QGIS, Mission Planner, ArduPilot SITL, Docker
Hardware	Raspberry Pi 5 (quad-core 2.4 GHz, 8 GB), Pixhawk 6C, GPS module, F450 frame, ESC + motors

Implementation Status

Component	Status	Evidence / Remarks
Literature review & problem formulation	Completed	3 papers surveyed; 3 gaps formalised; real-world evidence added
System design & architecture	Completed	Two-tier architecture, NAMOA* algorithm, and algebraic bridge designed; diagrams produced
Airspace graph construction	In Progress	Actively building the airspace graph model and 3-objective edge-cost assignment for the target delivery region
Tier 1: Multi-Objective A* (NAMOA*) engine	Planned	NAMOA* with 3-objective cost vectors + label-dominance pruning specified; implementation scheduled for Phase II
Tier 2: Payload-driven selector	Planned	Algebraic bridge and utility-scoring $\arg\max$ selection formalised; coding scheduled for Phase II
Benchmarking & evaluation	Planned	Metrics defined (HV, Δ, IGD, selection accuracy, runtime); benchmark harness vs. single-obj. A* and MOPSO planned for Phase II

Expected Results and Performance Targets

Payload Preset Weight Derivation

Preset	w_{risk}	w_{dist}	w_{energy}
Fragile medical (F=5, W=1, U=1)	0.667	0.200	0.133
Emergency medicine (F=1, W=1, U=3)	0.144	0.712	0.144
Disaster relief (F=1, W=3, U=1)	0.182	0.273	0.545
Mixed cargo (F=3, W=2, U=2)	0.375	0.375	0.250

Only emergency medicine selects an extreme Pareto point. **All other presets select non-extreme interior points** no single-objective baseline can produce.

Performance Targets

- HV: $\geq$20% higher than single-obj. A*
- Pareto front: 5–20 non-dominated routes per scenario
- Tier 2 selection latency: $<$50 ms on RPi 5
- Tier 2 accuracy: correct priority objective ranked highest across all 4 presets
- Hardware: ArduPilot SITL + outdoor delivery flight

Testing and Validation Plan

Test ID	Test Scenario	Expected Result	Status
TC-01	Valid payload tags (F, W, U, P_{lvl}) input	Correct $w_{\text{risk}}, w_{\text{dist}}, w_{\text{energy}}$ derived; weights sum to 1.0	Planned
TC-02	NAMOA* on 50/200/500-node graphs (S1/S2/S3)	Pareto front with zero dominated routes; HV $>$ single-obj. A*	Planned
TC-03	High fragility (F=5) vs. emergency medicine (U=3)	Significantly different non-extreme routes selected from same front	Planned
TC-04	Disconnected graph (no feasible path)	Graceful error with clear user message	Planned
TC-05	RPi 5 MAVLink mission upload to Pixhawk via SITL	Drone accepts mission and executes flight in ArduPilot SITL	Planned

Validation Method

Unit tests (pytest), MOPSO benchmarking (30 runs, swarm = 100, 200 iterations), ArduPilot SITL integration tests, and outdoor UAV flight tests with GPS telemetry logs.

Project Timeline (Jul 2026 – Mar 2027)

Activity	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar
Literature survey & formulation									
System design & architecture									
Tier 1: Multi-Objective A*									
Tier 2: Payload-Driven Selector									
Software simulation & benchmarking									
Hardware integration (RPi 5 + Pixhawk)									
Outdoor flight tests									
Documentation & report writing									

Review I: August 2026 **Final Submission:** March 2027

Team Responsibilities

Team Member	Responsibilities	Current Status
Thirumurugan RA	System architecture design, Tier 1 NAMOA* engine implementation, airspace graph construction (OSMnx), edge cost vectorization	In Progress
Meghna Manimaran	Algebraic bridge design, Tier 2 payload selector implementation, benchmarking suite development, documentation	In Progress
Shrika SP Thota	Literature survey & problem formulation, risk-cost model design, benchmarking methodology, Pareto-front visualization and result analysis	In Progress

Challenges, Risks, and Mitigation

Risk / Challenge	Possible Impact	Mitigation
Graph complexity (large OSM urban network)	Slow Pareto front construction	Spatial pruning, NAMOA* node-level label-dominance check, epsilon-dominance to limit front size
Hardware failure (RPi / Pixhawk)	Flight tests delayed	ArduPilot SITL validation first; spare hardware budget
Pareto front size too large	High memory and selection latency	Apply epsilon-pruning; front bounded to $\leq$100 routes
MAVLink integration issues	Mission upload failures	Validate protocol end-to-end in SITL before real drone flights

Conclusion and Future Work

Conclusion

This project addresses the critical gap of payload-blind UAV routing by proposing a two-tier payload-aware pipeline. Tier 1 builds a Pareto-optimal route set via NAMOA*; Tier 2 selects the mission route using an algebraic bridge that maps fragility, urgency, and payload weight to routing objective weights. Utility scoring over the full Pareto front replaces context-blind diversity criteria, enabling non-extreme route selection for mixed payloads that no single-objective baseline can produce.

Future Work

- Outdoor flight validation on a real RPi 5 + Pixhawk drone in a simulated urban delivery environment.
- Extend to multi-drone coordination with shared Pareto fronts.
- Incorporate real-time weather and dynamic obstacle data for in-flight re-routing with adaptive front refresh thresholds.

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Thank You

Questions and Discussion

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